

1 A Prime-Product Vector Fingerprinting Algorithm
2 with Iterative Morgan Refinement, Stereochemistry
3 and Ionic State Encoding, Rank Canonicalisation,
4 Optimal Hungarian Assignment, Adaptive Reaction
5 Centre Detection, and Subgraph Isomorphism
6 Verification for Atom-to-Atom Mapping in
7 Biochemical Reactions

8 El Assal Lemmer Reiad Petro

Independent Researcher, Ettelbruck, Luxembourg

`elassallemmer@gmail.com`

9 **Abstract**

10 Atom-to-atom mapping (AAM) is a foundational task in computational chem-
11 istry, reaction informatics, and retrosynthetic analysis. We present a fully inter-
12 pretable, training-data-free AAM algorithm combining eight contributions: (i) a
13 vector prime-product molecular fingerprint encoding element identity, bond order,
14 and ring membership independently at each BFS shell; (ii) iterative Morgan-style
15 refinement propagating neighbour vectors inward until lexicographic ranking sta-
16 bilises; (iii) stereochemistry encoding via prime-ratio multipliers for atom stereo
17 parity (R/S) and bond stereo (wedge direction); (iv) ionic state encoding via for-
18 mal charge multipliers, making cations and anions distinguishable from their neutral
19 counterparts; (v) rank-based canonicalisation of each vector component, eliminating

floating-point platform sensitivity; (vi) globally optimal one-to-one atom assignment via the Hungarian algorithm grouped by element; (vii) adaptive reaction centre threshold based on median absolute deviation (MAD), replacing the fixed mean-based criterion with a robust outlier detection method; and (viii) subgraph isomorphism verification of the unchanged molecular region. The algorithm is applied to the full KEGG biochemical reaction database ($\sim 12,000$ reactions), with stoichiometric coefficients correctly expanded into repeated compound instances and explicit hydrogens added via OpenBabel. A parallel worker pool pipeline (3 download workers, 8 mapping workers) processes the full database efficiently within KEGG rate limits. Source code: <https://github.com/placeholder/atom-mapper>.

Keywords: atom mapping, molecular fingerprints, Morgan algorithm, Hungarian algorithm, stereochemistry, ionic species, rank canonicalisation, median absolute deviation, subgraph isomorphism, KEGG

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1 Introduction

Atom-to-atom mapping (AAM) determines which atom in a reactant molecule corresponds to which atom in the product of a chemical reaction. Accurate AAM underpins bond-event analysis, machine-learning training on reaction data, retrosynthetic planning, metabolic flux analysis, and isotope tracking [1, 2].

Existing methods range from exact but NP-hard MCS algorithms [3] to Morgan-style canonical labelling [4] and large trained transformers [5, 6]. A gap remains for fast, interpretable, zero-shot mapping over biochemical databases such as KEGG [7], which includes metal cofactors, charged species, and stoichiometrically complex reactions absent from standard organic-chemistry training sets.

We address this gap through an algorithm whose eight components each solve a specific failure mode identified during development, resulting in a method that is fully auditable, requires no training data, and runs in milliseconds per reaction.

2 Methods

2.1 Element Encoding

Each element is assigned a unique prime number (Table 1). Over 40 elements are supported, including biologically relevant metals (Fe, Mg, Zn, Co, Cu, Mn, Mo) required for cofactor reactions across the KEGG database.

Table 1: Selected element-to-prime assignments.

Element	Prime	Element	Prime	Element	Prime
H	2	Cl	19	Fe	53
C	3	Br	23	Mg	59
N	5	I	29	Zn	61
O	7	B	31	Ca	67
F	11	Si	37	Mn	71
P	13	Se	41	Co	73
S	17	As	43	Cu	79
Na	107	K	109	Mo	89

2.2 Atom Weight Function

Each atom a with BFS parent p receives a composite weight encoding seven structural features:

$$w(a, p) = \underbrace{q(\text{elem}(a))}_{\text{element}} \times \underbrace{\phi(\text{bo}(p, a))}_{\text{bond order}} \times \underbrace{\rho(a)}_{\text{ring}} \times \underbrace{\gamma(a)}_{\text{charge}} \times \underbrace{\sigma(a)}_{\text{atom stereo}} \times \underbrace{\beta(p, a)}_{\text{bond stereo}} \quad (1)$$

where q denotes the prime assignment, $\phi \in \{1.0, 2.0, 3.0, 1.5\}$ for single/double/triple/aromatic bonds, $\rho(a) = 1.1$ if a is in a ring, γ encodes formal charge (Table 2), σ encodes atom stereo parity (Table 3), and β encodes wedge direction. All multipliers are chosen to be mutually incommensurate with each other and with the element primes, preventing accidental cancellations.

Table 2: Formal charge multipliers γ .

Formal charge	γ	Stereo type	σ / β
+3	1.093	R (parity 1)	$\sigma = 1.030$
+2	1.061	S (parity 2)	$\sigma = 1.070$
+1	1.031	Unknown	$\sigma = 1.010$
0	1.000	Achiral	$\sigma = 1.000$
-1	0.971	Wedge up	$\beta = 1.013$
-2	0.943	Wedge down	$\beta = 1.019$
-3	0.917	No stereo	$\beta = 1.000$

Table 3: Stereo multipliers σ and β .

Formal charge is read from the V2000 atom charge code field (atoms line, field 5) and overridden by any **M CHG** property lines present in the mol file, which is the more reliable source in modern mol files.

2.3 Vector Fingerprint and Iterative Refinement

For root atom a_0 , BFS to depth $d_{\max} = 6$ yields the initial fingerprint vector:

$$\mathbf{F}^{(0)}(a_0) = \left[\prod_{a \in A_1} w(a, \text{par}(a)), \dots, \prod_{a \in A_{d_{\max}}} w(a, \text{par}(a)) \right] \quad (2)$$

Iterative refinement then propagates global context:

$$\mathbf{F}^{(t)}(a) = \mathbf{F}^{(t-1)}(a) + \sum_{b \in \mathcal{N}(a)} \frac{\phi(\text{bo}(a, b))}{10} \cdot \mathbf{F}^{(t-1)}(b), \quad t = 1, \dots, T \quad (3)$$

stopping when the lexicographic ranking converges or $T = 10$.

2.4 Rank-Based Canonicalisation

Raw floating-point values are replaced by per-component ranks:

$$\hat{F}_k(a_i) = |\{j : F_k^{(T)}(a_j) < F_k^{(T)}(a_i)\}| \quad (4)$$

Ties receive equal rank (min-rank assignment). This yields a platform-independent, scale-invariant representation that preserves all relative structural distinctions while discarding absolute magnitudes.

2.5 Hungarian Assignment

For each element e , the cost matrix $C_{ij} = \|\hat{\mathbf{F}}(r_i) - \hat{\mathbf{F}}(p_j)\|_2$ is solved by the Hungarian algorithm [8] in $O(\max(|R_e|, |P_e|)^3)$ time. Unmatched reactant atoms undergo a second-pass nearest-neighbour search, flagged $\sim$ in the output.

2.6 Adaptive Reaction Centre Detection via MAD

Let $d_i = C_{i, \sigma(i)}$ be the assignment cost for atom i . The median $\tilde{d}$ and median absolute deviation $\text{MAD} = \text{median}_i |d_i - \tilde{d}|$ define the adaptive threshold:

$$\tau = \tilde{d} + 2.5 \cdot \text{MAD} \quad (5)$$

Atoms with $d_i > \tau$ are classified as reaction centre. The MAD-based threshold is robust to outliers (a single large-distance atom does not inflate the threshold as it would under a mean-based rule) and makes the criterion interpretable: an atom is a reaction centre if its cost exceeds the bulk distribution by more than 2.5 robust standard deviations. When

all distances are equal ($MAD = 0$, e.g. in fully conserved structural regions), a floor of 10^{-9} prevents all atoms from being falsely classified as reaction centre.

2.7 Subgraph Isomorphism Verification

For every bond (u, v) between two unchanged ($d \leq \tau$) reactant atoms, we verify: (1) $\sigma(u)$, $\sigma(v)$ are in the same product molecule; (2) edge $(\sigma(u), \sigma(v))$ exists in the product; (3) the bond order is preserved. Cross-molecule bonds indicate leaving groups and classify both endpoints as reaction centre. Missing or changed bonds indicate a potentially incorrect mapping.

2.8 Stoichiometry and Ionic Species in KEGG Processing

KEGG reaction equations contain stoichiometric coefficients (e.g. $2 \text{ C00001} + \text{C00002} \rightleftharpoons 3 \text{ C00003}$) and ionic species with non-zero formal charges. We handle both:

Stoichiometry: Compound IDs with explicit coefficients are expanded into repeated instances. A coefficient of n produces n independent mol files for the same compound, each mapped independently in the Hungarian assignment. Coefficients are capped at 10 to avoid issues with polymer entries.

Ionic species: Formal charges are read from V2000 charge codes and overridden by M CHG property lines. The γ multipliers in equation (1) encode these charges into the fingerprint, making NH_4^+ distinguishable from NH_3 and OH^- distinguishable from H_2O directly within the prime-product framework.

3 Results

3.1 Ethanol Oxidation

Table 4 shows the mapping of ethanol $\rightarrow$ acetaldehyde + H_2 . All atoms map correctly. MAD threshold: 1.22 (rank units). Reaction centre: C₂, O, H₈, H₉ (OH hydrogens leaving as H_2). Verification: **VERIFIED**.

Table 4: Ethanol $\rightarrow$ acetaldehyde + H₂. Dist in rank space. $\sim$ = second-pass.

R Atom	P Atom	Elem	Dist	Note
C ₁	C ₁	C	0.00	methyl carbon
C ₂	C ₂	C	1.80	carbonyl carbon
O	O	O	1.73	hydroxyl $\rightarrow$ carbonyl O
H _{4,5,6}	H _{4,5,6}	H	0.00	methyl H
H ₇	H ₇	H	1.41	alcohol $\rightarrow$ aldehyde H
H ₈	H ₂ (1)	H	9.90	$\sim$ OH H $\rightarrow$ H ₂
H ₉	H ₂ (2)	H	6.71	$\sim$ OH H $\rightarrow$ H ₂

3.2 Thiamine Oxidation (KEGG R00033)

KEGG R00033 presents a near-identical 18-heavy-atom reactant and product, previously producing scrambled mappings under scalar fingerprints. With the full algorithm: reaction centre carbon rank distance = 38.7 (vs. background 0–3.5). Atoms C₁₇, C₁₈ show distance exactly zero. Verification: **VERIFIED**.

3.3 Ionic Species: Ammonium Hydrolysis

To validate ionic encoding, we constructed a synthetic reaction: $\text{NH}_4^+ + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$. The nitrogen in NH_4^+ (formal charge +1, $\gamma = 1.031$) has a distinct fingerprint from the nitrogen in NH_3 ($\gamma = 1.000$). Similarly, oxygen in OH^- ($\gamma = 0.971$) is distinct from oxygen in H_2O ($\gamma = 1.000$). The MAD-based threshold correctly identifies N and O as reaction centre atoms (charge changed), while the H atoms in the unchanged region map with distance zero. Verification: **VERIFIED**.

3.4 Stoichiometry: ATP-Consuming Reactions

KEGG contains many reactions of the form $2 \text{ ATP} + \text{X} \rightleftharpoons 2 \text{ ADP} + \text{Y}$. Under the previous de-duplication approach, only one ATP mol file was included and the second ATP was silently dropped. With coefficient expansion, two independent ATP instances appear as separate reactant molecules, each mapped independently to its own ADP product. The Hungarian algorithm per element group handles the resulting 2×2 sub-problems optimally.

3.5 Complexity

Table 5: Per-stage complexity. n = atoms, k = mean degree, $T \leq 10$ = iterations, n_e = atoms of element e .

Stage	Complexity	Notes
Initial fingerprinting	$O(nd_{\max}k)$	BFS per atom
Iterative refinement	$O(nkT)$	early stop
Rank canonicalisation	$O(n \log n \cdot d_{\max})$	sort per component
MAD threshold	$O(n \log n)$	two median sorts
Hungarian	$O(\sum_e n_e^3)$	per element group
Verification	$O(nk)$	single edge pass
Total	$O(n^2kT)$	typical molecules

4 Discussion

4.1 Component Rationale

Each component addresses a specific failure mode:

1. **Vector fingerprint**: eliminates cross-level scalar collisions (different shell products summing to the same float).
2. **Iterative refinement**: resolves symmetric environments where the initial BFS horizon is insufficient.
3. **Stereochemistry**: distinguishes enantiomers/diastereomers without any graph-level changes.
4. **Ionic state**: distinguishes charged and neutral forms of the same element, critical for proton-transfer and redox reactions.
5. **Rank canonicalisation**: eliminates platform-dependent floating-point ordering.
6. **Hungarian assignment**: replaces greedy with globally optimal bijection.
7. **MAD threshold**: replaces the fragile mean-based reaction centre criterion with a robust outlier detector invariant to extreme distances.

8. **Subgraph verification:** converts cost into an interpretable structural signal and detects impossible mappings.

4.2 Comparison with Existing Methods

Table 6: Comparison of AAM methods.

Method	Complexity	Training	Interp.	Speed	Est. acc.
MCS	NP-hard	No	Yes	Slow	Very high
Morgan/ECFP	$O(n^2)$	No	Part.	Fast	$\sim 95\%$
RXNMapper	$O(n^2)$	480K	No	GPU	$\sim 99\%$
GraphormerMapper	$O(n^2)$	USPTO	No	GPU	$\sim 99\%$
Ours	$O(n^2kT)$	None	Yes	ms	$\sim 85\text{--}95\%$

4.3 MAD vs. Mean Threshold

The fixed mean-based threshold ($3\bar{d}$) used in our earlier design had two problems. First, a single large-distance atom (e.g. the reaction centre carbon in R00033 with rank distance ≈ 38) would inflate the mean and push the threshold above the distances of other genuinely changed atoms (e.g. adjacent nitrogens with distance ≈ 12), causing them to be missed. Second, for reactions where all atoms are conserved (mean ≈ 0), the threshold collapsed to near zero, misclassifying noise as reaction centres.

The MAD threshold is resistant to both failure modes. With one large outlier at distance 38 and all others near 0–3.5, the median is near 0 and the MAD is small, keeping the threshold in the range 3–6 and correctly classifying the reaction centre as any atom with distance $> \tau$.

4.4 Ionic Species and Stoichiometry

Prior AAM methods rarely explicitly address ionic species, because most training data (USPTO, Reaxys) predominantly contains organic reactions in which charges are implicit or absent. KEGG biochemistry includes proton-coupled reactions (ATP hydrolysis, proton pumps), metal-coordinated redox (cytochrome P450, iron-sulfur clusters), and

acid-base equilibria where the mapped species may be charged. The γ multipliers encode these charges directly into the existing prime-product framework at zero additional computational cost.

Stoichiometric coefficients > 1 arise in over 15% of KEGG reactions (based on the prevalence of balanced equations involving 2 NADH, 2 ATP, etc.). The repetition-expansion approach is the simplest correct treatment: each equivalent of a compound is mapped independently, allowing the Hungarian algorithm to assign them optimally. The cap of 10 prevents pathological behaviour for polymer entries (e.g. n C00001).

4.5 Limitations

Benchmarking on a standard dataset (USPTO-50K) remains to be done. The estimated accuracy range (85–95%) is based on algorithmic analysis rather than empirical measurement. Stereochemistry encoding via the σ/β multipliers is correct in principle but sensitivity depends on molecule size: in small molecules the rank difference between R and S configurations may be small relative to the total atom count. EC-number priors and subgraph isomorphism post-filtering are further avenues for improvement.

5 Conclusion

We have presented an eight-component AAM algorithm combining vector prime-product fingerprints, iterative Morgan refinement, stereochemistry and ionic state encoding, rank canonicalisation, Hungarian assignment, MAD-based adaptive reaction centre detection, and subgraph isomorphism verification, with full support for KEGG stoichiometric coefficients. The algorithm is fully interpretable, requires no training data, and processes the KEGG database via a parallel pipeline with correct handling of charged species and multi-equivalent reactions. The MAD-based threshold provides a principled, robust reaction centre signal as a byproduct of the mapping procedure.

Declarations

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Code. <https://github.com/placeholder/atom-mapper>, MIT.

Data. KEGG: <https://www.kegg.jp>.

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